

EXPERIMENT 1

SETTING UP PYTHON ENVIRONMENT AND LIBRARIES – JUPYTER NOTEBOOK

AIM:

To install Python, configure Jupyter Notebook, understand the notebook environment, execute Python programs, create Markdown cells, and use basic Jupyter Notebook features for Data Analytics applications.

ALGORITHM 1 :

Step 1: Download Python from
python.org Step 2: Run the installer
Step 3: Select "Add Python to
PATH" Step 4: Click Install Now
Step 5: Verify installation using:
python --version
Step 6: Observe installed Python version.

PROGRAM 1 :

Simple Python Program

```
[1]: print("Welcome to Data Analytics Lab")  
Welcome to Data Analytics Lab
```

PROGRAM 2:

Arithmetic Operations

```
[3]: a = 20  
b = 10  
  
print("Addition =", a+b)  
print("Subtraction =", a-b)  
print("Multiplication =", a*b)  
print("Division =", a/b)  
  
Addition = 30  
Subtraction = 10  
Multiplication = 200  
Division = 2.0
```

PROGRAM 3:

Display Student Details

```
name = "John"
dept = "AI&DS"
cgpa = 8.9

print("Name:", name)
print("Department:", dept)
print("CGPA:", cgpa)
```

Name: John
Department: AI&DS
CGPA: 8.9

PROGRAM 4:

Using NumPy Library

```
import numpy as np
a = np.array([10,20,30,40,50])
print(a)
print("Mean =", np.mean(a))
print("Sum =", np.sum(a))
```

[10 20 30 40 50]
Mean = 30.0
Sum = 150

PROGRAM 5:

Using Pandas Library

```
import pandas as pd
data = {
    'Name': ['John', 'David', 'Alex'], 'Marks': [85, 90, 95]
}

df = pd.DataFrame(data)
print(df)
```

	Name	Marks
0	John	85
1	David	90
2	Alex	95

PROGRAM 6:

Data Visualization

```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,15,25,30]

plt.plot(x,y)
plt.title("Sample Line Graph")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```

